

RAID (Redundant Array of Independent Disks) Configurations
Types of RAID: RAID 0, 1, 5, 10, and their use cases.
Importance of data redundancy.
Practical (2 hours): Simulate RAID configurations using virtual environments.
Set up RAID 1 for data mirroring.

Tasks

1. Create a comparison table listing RAID 0, RAID 1, RAID 5, and RAID 10, including columns for minimum number of disks, data redundancy, speed, and a real-world use case for each (such as Instagram photo storage, YouTube video servers, etc.).

Answer : -

RAID	Minimum Disks	Data Redundancy	Speed	Real-World Use Case
RAID 0	2	No	Very Fast	Video editing or temporary data
RAID 1	2	Yes, data is copied on both disks	Fast	Important files and small business servers
RAID 5	3	Yes, can survive 1 disk failure	Good	Photo storage servers like Instagram
RAID 10	4	Yes, combines mirroring and striping	Very Fast	Database servers and high-traffic websites like YouTube

2. Explain with your own example how RAID 1 provides data redundancy, using a scenario like storing WhatsApp chat backups on two disks, and describe what happens if one disk fails.

Answer :- RAID 1 stores the same data on two disks.

Example: Suppose WhatsApp chat backups are stored on Disk 1. RAID 1 automatically makes a copy of the same backup on Disk 2.

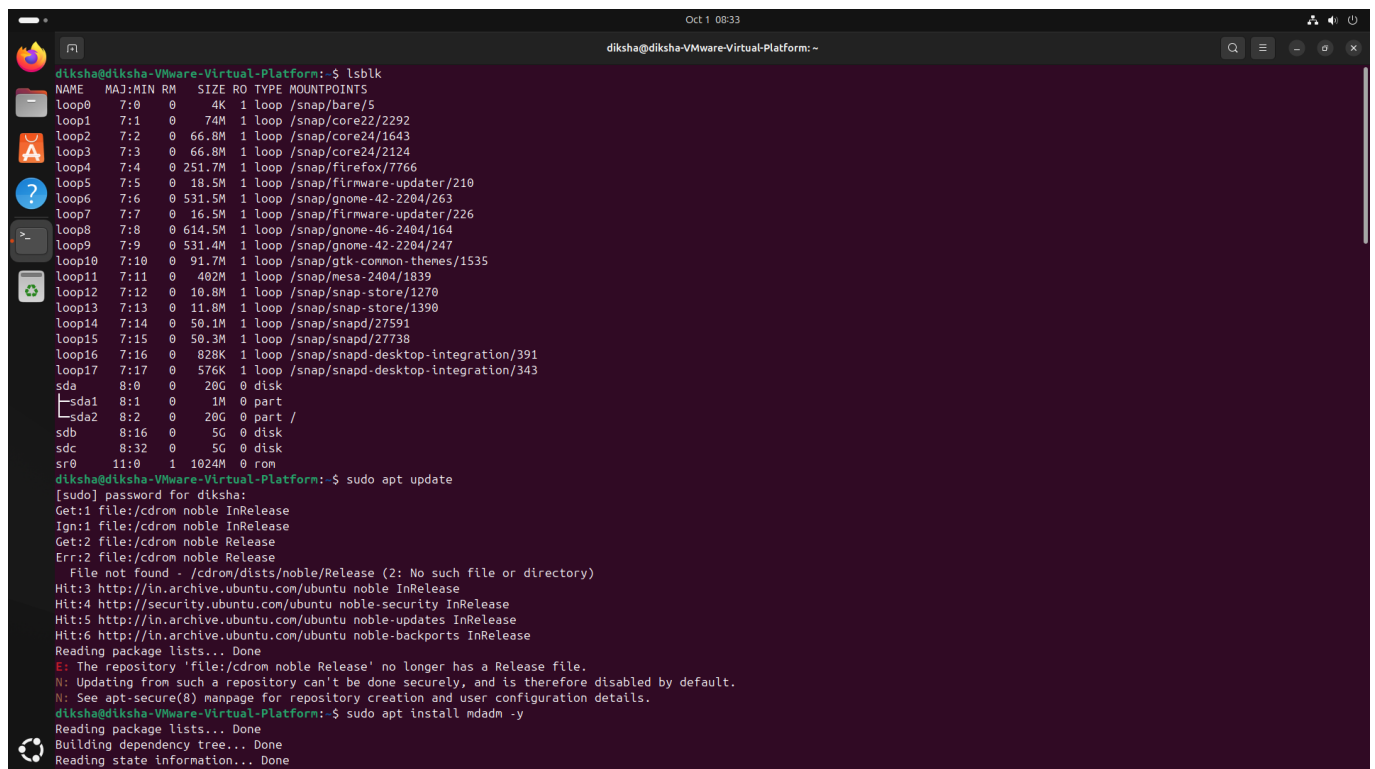
If Disk 1 fails, the data is still available on Disk 2, so the chat backups are not lost. After replacing the failed disk, the data can be copied to the new disk.

Therefore, RAID 1 provides data protection by keeping a duplicate copy of the data on another disk.

3. Simulate a RAID 1 setup using VirtualBox and Ubuntu: create two virtual hard disks, use the 'mdadm' tool to configure them in RAID 1, and take a screenshot of the output from the command 'cat /proc/mdstat' showing the RAID array.

Hint: Search for 'mdadm RAID 1 Ubuntu VirtualBox' for step-by-step guides.

Answer :-



```
diksha@diksha-VMware-Virtual-Platform:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
loop0       7:0      0   4K  1 loop  /snap/bare/5
loop1       7:1      0   74M  1 loop  /snap/core22/2292
loop2       7:2      0  66.8M  1 loop  /snap/core24/1643
loop3       7:3      0  66.8M  1 loop  /snap/core24/2124
loop4       7:4      0 251.7M  1 loop  /snap/firefox/7766
loop5       7:5      0  18.5M  1 loop  /snap/firefox-updater/210
loop6       7:6      0 531.5M  1 loop  /snap/gnome-42-2204/263
loop7       7:7      0  16.5M  1 loop  /snap/firmware-updater/226
loop8       7:8      0 614.5M  1 loop  /snap/gnome-46-2404/164
loop9       7:9      0 531.4M  1 loop  /snap/gnome-42-2204/247
loop10      7:10     0  91.7M  1 loop  /snap/gtk-common-themes/1535
loop11      7:11     0  402M  1 loop  /snap/mesa-2404/1839
loop12      7:12     0  10.8M  1 loop  /snap/snap-store/1270
loop13      7:13     0  11.8M  1 loop  /snap/snap-store/1390
loop14      7:14     0  50.1M  1 loop  /snap/snapd/27591
loop15      7:15     0  50.3M  1 loop  /snap/snapd/27738
loop16      7:16     0   828K  1 loop  /snap/snapd-desktop-integration/391
loop17      7:17     0   576K  1 loop  /snap/snapd-desktop-integration/343
sda         8:0      0   20G  0 disk
├─sda1      8:1      0    1M  0 part
└─sda2      8:2      0   20G  0 part /
sdb         8:16     0    5G  0 disk
sdc         8:32     0    5G  0 disk
sr0        11:0     1 1024M  0 rom

diksha@diksha-VMware-Virtual-Platform:~$ sudo apt update
[sudo] password for diksha:
Get:1 file:/cdrom/noble InRelease
Ign:1 file:/cdrom/noble InRelease
Get:2 file:/cdrom/noble Release
Err:2 file:/cdrom/noble Release
  File not found - /cdrom/dists/noble/Release (2: No such file or directory)
Hit:3 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
E: The repository 'file:/cdrom/noble Release' no longer has a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
diksha@diksha-VMware-Virtual-Platform:~$ sudo apt install mdadm -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
mdadm is already installed and up-to-date.
```


4. Delete a file from your RAID 1 array, then power off one of the virtual disks (simulate a disk failure), and check if the file is still accessible from the RAID array.

Hint: This tests if RAID 1 is really mirroring your data.

Answer :-

```
diksha@diksha-VMware-Virtual-Platform:~$ cat /proc/mdstat
Personalities : [raid0] [raid1] [raid4] [raid5] [raid6] [raid10] [linear]
md127 : active (auto-read-only) raid1 sdb[0] sdc[1]
5237760 blocks super 1.2 [2/2] [UU]

unused devices: <none>

diksha@diksha-VMware-Virtual-Platform:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0     4K  1 loop /snap/bare/5
loop1       7:1      0    74M  1 loop /snap/core22/2292
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loop9       7:9      0  614.5M  1 loop /snap/gnome-46-2404/164
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loop12      7:12     0   10.8M  1 loop /snap/snap-store/1270
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loop15      7:15     0   50.3M  1 loop /snap/snapd/27738
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loop17      7:17     0   576K  1 loop /snap/snapd-desktop-integration/343
sda         8:0      0    20G  0 disk
├─sda1      8:1      0     1M  0 part
├─sda2      8:2      0    20G  0 part /
sdb         8:16     0     5G  0 disk
├─md127     9:127    0     5G  0 raid1
sdc         8:32     0     5G  0 disk
├─md127     9:127    0     5G  0 raid1
sr0         11:0     1  1024M  0 rom

diksha@diksha-VMware-Virtual-Platform:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            387M  2.2M  385M   1% /run
/dev/sda2        20G   13G   6.1G  68% /
tmpfs            1.9G   0  1.9G   0% /dev/shm
tmpfs            5.0M   8.0K   5.0M   1% /run/lock
tmpfs            387M  116K  387M   1% /run/user/1000

diksha@diksha-VMware-Virtual-Platform:~$ sudo mkdir -p /mnt/raid1
[sudo] password for diksha:
diksha@diksha-VMware-Virtual-Platform:~$ sudo mdadm --readwrite /dev/md127
diksha@diksha-VMware-Virtual-Platform:~$ sudo mount /dev/md127 /mnt/raid1
```

```
├─sda2      8:2      0    20G  0 part /
sdb         8:16     0     5G  0 disk
├─md127     9:127    0     5G  0 raid1
sdc         8:32     0     5G  0 disk
├─md127     9:127    0     5G  0 raid1
sr0         11:0     1  1024M  0 rom

diksha@diksha-VMware-Virtual-Platform:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            387M  2.2M  385M   1% /run
/dev/sda2        20G   13G   6.1G  68% /
tmpfs            1.9G   0  1.9G   0% /dev/shm
tmpfs            5.0M   8.0K   5.0M   1% /run/lock
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diksha@diksha-VMware-Virtual-Platform:~$ sudo mkdir -p /mnt/raid1
[sudo] password for diksha:
diksha@diksha-VMware-Virtual-Platform:~$ sudo mdadm --readwrite /dev/md127
diksha@diksha-VMware-Virtual-Platform:~$ sudo mount /dev/md127 /mnt/raid1
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tmpfs            387M  2.2M  385M   1% /run
/dev/sda2        20G   13G   6.1G  68% /
tmpfs            1.9G   0  1.9G   0% /dev/shm
tmpfs            5.0M   8.0K   5.0M   1% /run/lock
tmpfs            387M  116K  387M   1% /run/user/1000
/dev/md127      4.9G   32K   4.6G   1% /mnt/raid1
diksha@diksha-VMware-Virtual-Platform:~$ ls -l /mnt/raid1
total 24
drwx----- 2 root root 16384 Oct  1 08:32 lost-found
-rw-r--r--  1 root root    17 Oct  1 08:33 survive.txt
-rw-r--r--  1 root root    12 Oct  1 08:41 test.txt
diksha@diksha-VMware-Virtual-Platform:~$ cat /mnt/raid1/test.txt
RAID 1 test
diksha@diksha-VMware-Virtual-Platform:~$ echo "RAID 1 test" | sudo tee /mnt/raid1/test.txt
RAID 1 test
diksha@diksha-VMware-Virtual-Platform:~$ cat /mnt/raid1/test.txt
RAID 1 test
diksha@diksha-VMware-Virtual-Platform:~$
diksha@diksha-VMware-Virtual-Platform:~$
diksha@diksha-VMware-Virtual-Platform:~$ cat /proc/mdstat
Personalities : [raid0] [raid1] [raid4] [raid5] [raid6] [raid10] [linear]
md127 : active raid1 sdb[0] sdc[1]
5237760 blocks super 1.2 [2/2] [UU]

unused devices: <none>
diksha@diksha-VMware-Virtual-Platform:~$
```

sudo shutdown now

remove one disk --- 5GB

```
Oct 1 08:52
diksha@diksha-VMware-Virtual-Platform: ~
loop10  7:10  0 531.5M  1 loop  /snap/gnome-42-2204/263
loop11  7:11  0 91.7M  1 loop  /snap/gtk-common-themes/1535
loop12  7:12  0 10.8M  1 loop  /snap/snap-store/1270
loop13  7:13  0 50.1M  1 loop  /snap/snapd/27591
loop14  7:14  0 11.8M  1 loop  /snap/snap-store/1390
loop15  7:15  0 576K  1 loop  /snap/snapd-desktop-integration/343
loop16  7:16  0 828K  1 loop  /snap/snapd-desktop-integration/391
loop17  7:17  0 50.3M  1 loop  /snap/snapd/27738
sda      8:0    0 20G    0 disk
├─sda1   8:1    0 1M     0 part
├─sda2   8:2    0 20G    0 part /
├─sdb    8:16   0 5G     0 disk
└─md127  9:127  0 0B     0 md
sr0      11:0   1 1024M   0 rom

diksha@diksha-VMware-Virtual-Platform:~$ sudo mdadm --stop /dev/md127
[sudo] password for diksha:
mdadm: stopped /dev/md127
diksha@diksha-VMware-Virtual-Platform:~$ sudo mdadm --assemble --run /dev/md127 /dev/sdb
mdadm: /dev/md127 has been started with 1 drive (out of 2).
diksha@diksha-VMware-Virtual-Platform:~$ cat /proc/mdstat
Personalities : [raid0] [raid1] [raid4] [raid5] [raid6] [raid10] [linear]
md127 : active raid1 sdb[1]
5237760 blocks super 1.2 [2/1] [_U]

unused devices: <none>
diksha@diksha-VMware-Virtual-Platform:~$ sudo blkid /dev/md127
/dev/md127: UUID="3ba44c4c-e332-496d-ad36-da1dea34900b" BLOCK_SIZE="4096" TYPE="ext4"
diksha@diksha-VMware-Virtual-Platform:~$ sudo mkdir -p /mnt/raid1
diksha@diksha-VMware-Virtual-Platform:~$ sudo mount /dev/md127 /mnt/raid1
diksha@diksha-VMware-Virtual-Platform:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           387M  2.2M  385M   1% /run
/dev/sda2       20G   13G   6.0G  68% /
tmpfs           1.9G   0 1.9G   0% /dev/shm
tmpfs           5.0M   8.0K  5.0M   1% /run/lock
tmpfs           387M  116K  387M   1% /run/user/1000
/dev/md127      4.9G   32K   4.6G   1% /mnt/raid1
diksha@diksha-VMware-Virtual-Platform:~$ ls -l /mnt/raid1
total 24
drwx----- 2 root root 16384 Oct 1 08:32 lost+found
-rw-r--r--  1 root root    17 Oct 1 08:33 survive.txt
-rw-r--r--  1 root root    12 Oct 1 08:45 test.txt
diksha@diksha-VMware-Virtual-Platform:~$ cat /mnt/raid1/test.txt
RAID 1 test
diksha@diksha-VMware-Virtual-Platform:~$
```

5. Use ChatGPT to ask: 'What are the main differences between RAID 5 and RAID 10 for a cloud storage service like Google Drive?' Summarize the answer in 3 bullet points in your own words.

Answer : -

RAID 5 vs RAID 10 :-

- **Data protection:** RAID 5 can survive **one disk failure**, while RAID 10 can usually survive **one or more disk failures** depending on which disks fail.
- **Performance:** RAID 10 generally provides **better read/write performance** than RAID 5, especially for frequent write operations.
- **Storage:** RAID 5 uses storage space more efficiently, while RAID 10 needs **more disks and more storage capacity** because data is mirrored.